Lakshya JEE 2.0 (2024)

Matrices

DPP-03

1. If $A=\begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, then x =

(1) 3 (2) 5
(3) 2 (4) 4

2. If A and B are two matrices such that both $A + B$ and $A \times B$ are defined, then
(1) A and B are of same order
(2) A is of order $m \times m$ and B is of order $n \times n$, $m \neq n$.
(3) Both A and B are of same order $n \times n$.
(4) A is of order $m \times n$ and B is of order $n \times m$

3. If A = $[a_{ij}]$ is a skew-symmetric matrix of order n, then a_{ii} is equal to:
(1) 0 for some i
(2) 0 for all $i = 1, 2, \dots, n$
(3) 1 for some i
(4) 1 for all $i = 1, 2, \dots\dots, n$

4. If a matrix A is both symmetric and skew-symmetric matrix, then:
(1) A is a diagonal matrix
(2) A is zero square matrix
(3) A is a square matrix
(4) None of the above

5. If A and B are symmetric matrices of the same order, then:
(1) AB is a symmetric matrix
(2) A – B is skew-symmetric matrix
(3) AB + BA is a symmetric matrix
(4) AB – BA is a symmetric matrix

6. The square matrix A = $[a_{ij}]$ given by $a_{ij} = (i - j)^3$ is a
(1) symmetric matrix
(2) skew-symmetric matrix
(3) diagonal matrix
(4) None of these

7. If $A=\begin{bmatrix} 6 & 8 & 5 \\ 4 & 2 & 3 \\ 9 & 7 & 1 \end{bmatrix}$ is the sum of a symmetric matrix B and skew –symmetric matrix C, then B is equal to

(1) $\begin{bmatrix} 6 & 6 & 7 \\ 6 & 2 & 5 \\ 7 & 5 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 0 & 2 & -2 \\ -2 & 5 & -2 \\ 2 & 2 & 0 \end{bmatrix}$

(3) $\begin{bmatrix} 6 & 6 & 7 \\ -6 & 2 & -5 \\ -7 & 5 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 0 & 6 & -2 \\ 2 & 0 & -2 \\ -2 & -2 & 0 \end{bmatrix}$

8. Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is:
(1) 6 (2) 4
(3) 1 (4) 12

9. If A is a 3 × 3 matrix whose elements are from the set {–1, 0, 1}. Find the number of matrices A such that $tr(AA^T) = 3$. Where tr(A) is sum of diagonal element of matrix A.
(1) 572 (2) 612
(3) 672 (4) 682

10. The maximum number of different elements required to form a symmetric matrix of order 6 is
(1) 15 (2) 17
(3) 19 (4) 21

11. If A is a skew-symmetric matrix of order 2 and B, C are matrices $\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$, $\begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$ respectively, then $A^3(BC) + A^5(B^2C^2) + A^7(B^3C^3) + \ldots + A^{2n+1}(B^nC^n)$, is

(1) a symmetric matrix
(2) a skew-symmetric matrix
(3) an identity matrix
(4) None of these

12. If $A = [a_{ij}]$ is a square matrix of even order that $a_{ij} = i^2 - j^2$, then

(1) A is a skew-symmetric matrix and $|A| = 0$
(2) A is symmetric matrix and $|A|$ is a square
(3) A is symmetric matrix and $|A| = 0$
(4) none of these

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)
2. (3)
3. (2)
4. (2)
5. (3)
6. (2)
7. (1)
8. (2)
9. (3)
10. (4)
11. (2)
12. (4)